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DISSERTATION REPORT

Liquidity Management In Indian MSMEs

A Comparative Analysis of Working-Capital Drivers in Manufacturing and Automobile Sectors

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Declaration

I hereby declare that the dissertation entitled "Liquidity Management in Micro, Small and Medium Enterprises " is submitted as part of PGDM curriculum.
This work is an original piece of research conducted by me. All data used in this study has been sourced from publicly available databases including the Ministry of MSME (Government of India), Reserve Bank of India, SIDBI, CMIE Prowess, MCA21, NSE and BSE listed MSME annual reports, NSSO, and other government and regulatory databases. All sources have been duly acknowledged both in the body of the report and in the bibliography.
This dissertation has not been submitted to any other university or institution for the award of any degree or diploma.

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This is to certify that the dissertation entitled "Liquidity Management in Micro, Small and Medium Enterprises" submitted by Ashish Chaudhari (Roll No. MNP20242016) is a bonafide record of research work carried out under my guidance. The work is original and has not been submitted elsewhere for any degree.

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Abstract

This dissertation empirically examines the determinants of short-term liquidity distress in Indian Micro, Small and Medium Enterprises (MSMEs), with a specific focus on comparing working-capital drivers between manufacturing and automobile/auto-component firms. Using a panel dataset of 54 listed MSME firms — 32 in manufacturing and 22 in the automobile sector — spanning seven financial years from FY2018–19 to FY2024–25 (378 firm-year observations sourced from the Screener database and NSE/BSE filings), the study employs sector-wise Ordinary Least Squares (OLS) regression, robustness checks using a continuous liquidity measure, and a pooled interaction model to investigate whether the primary driver of liquidity distress differs across sectors.

The empirical results confirm partial sectoral heterogeneity. In the manufacturing sector, Days Sales Outstanding (DSO) emerges as a statistically significant predictor of the binary distress dummy ($p \approx 0.054$, borderline), while Return on Assets (ROA) is the dominant control variable ($p < 0.001$). In the automobile sector, Inventory Days (DIO) attains significance under the continuous current-ratio specification (coefficient: +0.0085; $p = 0.012$), confirming the operational inventory-intensity of auto-component firms. Robustness checks using current ratio as a continuous dependent variable corroborate the primary findings. The pooled interaction term (DIO x Sector dummy) does not reach conventional significance ($p = 0.319$), indicating that while sector-level patterns differ directionally, differences in marginal effects are not statistically distinguishable at the pooled level after controlling for firm characteristics.

These findings carry important implications for MSME managers — manufacturing firms should prioritise receivables management as the first line of liquidity defence, while automobile firms must address inventory efficiency — and for policymakers designing liquidity-support interventions, credit guarantees, and payment-timing reforms for India's MSME sector.

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Chapter 1: Introduction

1.1 Background and Context

Micro, Small and Medium Enterprises (MSMEs) form an indispensable pillar of India's industrial and employment landscape. These enterprises contribute substantially to manufacturing output, export earnings, and employment generation, functioning as critical nodes in both domestic and international supply chains. According to official disclosures from the Ministry of Micro, Small and Medium Enterprises, MSME-related products accounted for approximately 45–46 per cent of India's total exports in FY2023–24, reflecting the macro-economic significance of this enterprise category. The Government of India's policy discourse has consistently emphasised the importance of enabling MSMEs to access formal financing, improve cash flows, and enhance operational resilience.

Liquidity — defined as a firm's capacity to meet short-term financial obligations without operational disruption — is arguably the most critical financial concern for MSMEs. Unlike large corporations with diversified funding sources, established credit relationships, and deep capital market access, MSMEs operate under considerably tighter financial constraints. Their liquidity is primarily determined by the management of working capital, which encompasses trade receivables (accounts receivable), inventories, and trade payables. The efficiency with which these three working-capital components are managed directly determines how quickly a firm can convert operational activities into cash.

For Indian MSMEs, liquidity stress has become a structural feature rather than a cyclical anomaly. Delayed payments from large buyers, inadequate access to formal bank credit due to collateral constraints, and vulnerability to demand shocks collectively impair the cash conversion cycle of smaller firms. The Reserve Bank of India has repeatedly identified cash-flow constraints as a key structural impediment to MSME growth, and regulatory bodies have introduced platforms such as the Trade Receivables Discounting System (TReDS) to address receivables-related liquidity gaps. The COVID-19 pandemic further exacerbated these constraints, causing acute short-term cash-flow stress across MSME sectors during FY2019–21, a period captured in this study's data.

1.2 Problem Statement

Despite the macroeconomic importance of MSMEs and the recognised centrality of liquidity management to their survival, there is limited empirical evidence that directly compares which working-capital component most powerfully explains liquidity distress across sectors, using consistent firm-level panel data and formal robustness tests. Existing Indian studies predominantly focus on the relationship between working capital and profitability, rather than liquidity distress per se, and rarely adopt comparative sectoral frameworks or employ interaction tests to formally assess sectoral heterogeneity. This study addresses these gaps by constructing an original panel dataset of listed MSME firms spanning seven financial years and applying a rigorous econometric framework.

1.3 Research Objectives

The primary objective of this dissertation is to empirically identify which working-capital components explain short-term liquidity distress in Indian MSMEs and whether the dominant driver differs between manufacturing and automobile/auto-component firms. The specific objectives are:

- To estimate the relationship between working-capital components (DSO, DIO, DPO) and liquidity distress among MSME firms across two sectors.
- To compare working-capital determinants of liquidity distress between manufacturing and automobile MSMEs, and to examine whether the primary driver differs by sector.
- To test the robustness of the findings using both binary (distress dummy) and continuous (current ratio) dependent variables, and to formally test sectoral heterogeneity using a pooled interaction model.

1.4 Research Questions and Hypotheses

The central research question of this study is: Do the working-capital drivers of liquidity distress differ between manufacturing and automobile MSMEs in India? Two sector-specific hypotheses are tested:

- H_0 (Manufacturing): Receivables management, as measured by Days Sales Outstanding (DSO), is the primary and statistically significant predictor of liquidity distress in manufacturing MSMEs.
- H_1 (Automobile): Inventory management, as measured by Inventory Days (DIO), is the primary and statistically significant predictor of liquidity distress in automobile MSMEs.

1.5 Scope of the Study

The study uses a panel dataset of 54 listed MSME firms — 32 in manufacturing and 22 in the automobile and auto-components sector — covering seven financial years from FY2018–19 to FY2024–25, yielding 378 firm-year observations after data cleaning, removal of zero-revenue observations, and winsorisation of extreme ratio values at the 1st and 99th percentiles. Financial data were sourced from the Screener financial database and verified against audited annual reports and NSE/BSE filings. The study is conducted under the institutional affiliation of the International School of Business and Media, under the supervision of Prof. Harikrishnan K.S.

1.6 Significance of the Study

This research carries relevance across three dimensions. First, from a policy perspective, understanding which working-capital levers most frequently precipitate liquidity distress enables policymakers to design more targeted interventions. Second, from a managerial perspective, the findings enable CFOs and treasurers of MSME firms to prioritise working-capital interventions based on their sectoral context rather than applying generic cash management principles. Third, the study contributes to academic literature by combining sector-wise regressions, robustness checks, and a formal interaction test in the Indian MSME context, an approach that is underrepresented in the existing literature.

1.7 Key Variables

The dependent variables are: (i) a binary distress dummy, coded 1 when the current ratio falls below unity, and (ii) the current ratio as a continuous measure for robustness. The independent variables are the three working-capital components: DSO (Days Sales Outstanding / Debtor Days), DIO (Days Inventory Outstanding / Inventory Days), and DPO (Days Payable Outstanding). Control variables include firm size (natural logarithm of total assets), leverage (total debt to total assets), and return on assets (EBIT divided by total assets).

Chapter 2: Literature Review

2.1 Theories of Liquidity Management

The theoretical foundation of liquidity management draws on several interrelated frameworks. The classical liquidity–profitability trade-off, articulated by Horne and Wachowicz (2005), posits that firms holding more liquid assets enjoy greater safety but sacrifice profitability, since liquid assets typically yield lower returns than productive capital. For MSMEs, however, this trade-off is asymmetric: given their higher vulnerability to cash-flow shocks and limited credit access, liquidity preservation often dominates profitability optimisation.

The pecking order theory (Myers and Majluf, 1984) suggests that firms prefer internal financing over external sources, implying that efficient working-capital management — converting assets to cash quickly — is especially important for MSMEs that may lack ready access to capital markets. The Cash Conversion Cycle (CCC) framework, originally developed by Richards and Laughlin (1980), is the central operational concept linking working capital to liquidity:

$$CCC = DSO + DIO - DPO$$

A longer CCC implies that capital is tied up in operations for extended periods, increasing liquidity pressure. Firms can reduce their CCC by accelerating collections (lower DSO), reducing inventory holding (lower DIO), or extending supplier payment terms (higher DPO). Each lever has sector-specific feasibility constraints, motivating the comparative analysis in this study.

2.2 Working Capital and Firm Performance: Empirical Evidence

A substantial body of empirical literature documents the relationship between working-capital management and firm outcomes. Deloof (2003), using Belgian firm data, found that reducing the CCC significantly improves profitability, with trade receivables and inventories the most critical components. Gill, Biger, and Mathur (2010) confirmed these findings in a North American context, demonstrating negative associations between receivable days, inventory days, and profitability.

In the Indian context, Sharma and Kumar (2011) examined the working capital–profitability relationship for a sample of Indian firms and found that receivable days negatively affect profitability, while DPO is positively associated with returns. Ganesan (2007) showed that firms with shorter cash conversion cycles exhibit superior financial performance. The study by Shroff (2013) on ITC Limited in the Indian food and beverages industry found that ROTA was not very sensitive to changes in working capital leverage, highlighting that sector-specific contexts moderate the working-capital–performance relationship. The working capital management analysis of cement units in Rajasthan (Alrjoub et al., 2012) showed significant differences in inventory size and receivables across companies, emphasising the importance of granular, sector-level analysis rather than broad generalisations.

Bardia (2004) specifically highlighted the role of receivables mismanagement as a key liquidity risk in smaller Indian firms. Patta (2023) found in a comparative study of Indian manufacturing and service sector companies that manufacturing firms carried higher DSO and leverage but also demonstrated superior ROA and ROE, suggesting that sector-level financial characteristics are systematically different. Most Indian studies, however, focus on profitability as the dependent variable rather than liquidity distress per se.

2.3 Liquidity Constraints in Indian MSMEs

Indian MSMEs face structural liquidity challenges that are qualitatively different from those confronting large enterprises. Delayed payments from large corporate buyers are among the most frequently cited challenges: payment cycles of 60–120 days or more are common in manufacturing supply chains, and the resultant receivables accumulation directly strains working capital. The Ministry of Micro, Small and Medium Enterprises has acknowledged delayed payments as a systemic problem and has established the Micro and Small Enterprises Facilitation Council (MSEFC) and the TReDS platform to address this issue.

Bank financing of working capital is another persistent constraint. MSMEs often lack the collateral and credit history required for formal bank loans, and their dependence on informal trade credit at higher implicit costs compounds their financial fragility. The Reserve Bank of India's Working Group on MSMEs (2019) has consistently highlighted access to affordable, timely working capital finance as a structural bottleneck. The comparative analysis of public sector banks by Attarwala (2025) further confirms that liquidity management is a complex, multi-variable challenge, with regression analysis demonstrating that liquidity ratios significantly impact financial performance indicators.

2.4 Sectoral Heterogeneity in Liquidity Drivers

Manufacturing MSMEs engaged in industrial goods, capital equipment, and specialised components typically extend long trade credit to buyers, resulting in elevated DSO. Buyer concentration risk — where a few large clients account for the bulk of revenues — amplifies receivables-related distress. The study by Chandrabai and Rao (2011) on Indian electrical equipment manufacturers (BHEL and ABB) found that debtor conversion periods dominated the operating cycle, confirming that receivables are the primary liquidity lever in capital goods manufacturing.

Automobile and auto-component firms, by contrast, operate in a capital-intensive, inventory-heavy environment. Multi-tier supply chains require firms to maintain substantial inventories of raw materials, sub-assemblies, and finished goods. The consequence is that DIO is structurally higher in automobile clusters, and inventory mismanagement — through overstocking, obsolescence, or production disruption — directly threatens liquidity.

2.5 Liquidity Distress vs. Profitability

Most Indian empirical literature treats working capital as a driver of profitability. Liquidity distress, by contrast, reflects operational vulnerability: the inability of a firm to cover its current liabilities with current assets signals an immediate threat to going concern, even if the firm is technically profitable over the medium term. The binary distress dummy (current ratio < 1) used in this study captures precisely this dimension of financial fragility. Altman (1968), in his foundational Z-Score model, included a current ratio term as a predictor of corporate distress. More recent work by Patel and Bhatt (2020) in the Indian MSME context identified low current ratios as the single most reliable leading indicator of firm-level distress in small enterprises.

2.6 Research Gaps

The preceding review identifies the following gaps that this study addresses: (i) limited empirical studies specifically examining liquidity distress in Indian MSMEs using firm-level panel data; (ii) insufficient sectoral comparison of working-capital determinants, with most studies conflating diverse MSME sub-sectors; (iii) sparse use of interaction models to formally test whether sector moderates the working-capital–distress relationship; and (iv) overemphasis on profitability rather than liquidity risk as the outcome of interest.

Chapter 3: Research Methodology

3.1 Research Design

This study employs a quantitative, causal research design using panel data from listed MSME firms. Panel data, which combines cross-sectional and time-series variation, is well suited to studying firm-level financial behaviour over time, enabling the researcher to exploit both cross-sectional differences between firms and temporal variation within firms. The empirical strategy encompasses descriptive analysis, inferential statistics (t-tests), correlation analysis, sector-wise OLS regression, robustness checks, and a pooled interaction regression to test formal sectoral heterogeneity.

3.2 Data Sources and Time Period

Secondary financial data were collected from the Screener financial database, supplemented by audited annual financial statements and NSE/BSE filings where necessary. Screener aggregates financial data from audited accounts filed by listed firms in India, providing consistent definitions of working-capital ratios, profitability measures, and balance-sheet items. The study period spans seven financial years from FY2018–19 to FY2024–25, capturing pre-pandemic, pandemic-period, and post-pandemic financial behaviour, thereby enhancing the generalisability of the findings.

3.3 Sample Selection

The final sample consists of 54 listed MSME firms: 32 in manufacturing and 22 in the automobile and auto-components sector, yielding 378 firm-year observations. Firms were selected based on the following criteria:

- Classification as MSMEs based on prevailing investment and turnover thresholds under the MSME Development Act.
- Availability of complete financial data for all key variables over the study period.
- Continuous listing and operation on the NSE or BSE during the study period.
- Exclusion of zero-revenue firm-years and observations with missing data for core working-capital variables.

The manufacturing sample includes 32 firms spanning engineering goods, process equipment, precision components, and industrial machinery. The automobile sample includes 22 firms engaged in auto-component manufacturing, including bearings, pistons, springs, castings, and specialty components.

3.4 Variable Construction

3.4.1 Dependent Variables

Distress Dummy: Binary variable coded 1 when Current Ratio < 1 (indicating short-term illiquidity), and 0 otherwise. Current Ratio (Robustness Variable): Defined as Current Assets divided by Current Liabilities. Used as a continuous liquidity measure to test whether binary-model results are robust to specification choice.

3.4.2 Independent Variables

Days Sales Outstanding (DSO): Number of days of revenue represented by outstanding trade receivables. Days Inventory Outstanding (DIO): Number of days of cost of goods sold represented by inventory held. Days Payable Outstanding (DPO): Number of days of cost of goods sold represented by outstanding trade payables.

3.4.3 Control Variables

Size: Natural logarithm of total assets. Leverage: Total debt divided by total assets. Return on Assets (ROA): EBIT divided by total assets.

3.5 Data Cleaning and Winsorisation

To ensure statistical reliability and mitigate the influence of data anomalies, the following cleaning steps were applied: (i) zero-revenue observations and firm-years with missing values in any core variable were removed; (ii) extreme values in ratio variables (DSO, DIO, DPO, Current Ratio) were winsorised at the 1st and 99th percentile thresholds to reduce outlier distortion while preserving the full sample size; and (iii) derived variables — Size, Leverage, ROA, and the Distress Dummy — were constructed from raw balance-sheet and income-statement data, following standard definitions.

3.6 Econometric Specifications

The primary econometric model for sector-specific distress regressions is:

$$Distress_{it} = \alpha + \beta_1 DSO_{it} + \beta_2 DIO_{it} + \beta_3 DPO_{it} + \beta_4 Size_{it} + \beta_5 Leverage_{it} + \beta_6 ROA_{it} + \varepsilon_{it}$$

This linear probability model is estimated separately for manufacturing and automobile sub-samples using OLS. For robustness, the dependent variable is replaced with Current Ratio (continuous). The pooled interaction model for formal sectoral heterogeneity testing is:

$$Distress_{it} = \alpha + \beta_1 DIO_{it} + \beta_2 DSO_{it} + \ldots + \beta_7 SectorDummy_{it} + \beta_8 (DIO \times SectorDummy)_{it} + \varepsilon_{it}$$

3.7 Limitations of Methodology

The methodology is subject to several limitations. First, the use of OLS on a binary dependent variable can produce predicted probabilities outside the [0,1] range. Second, the panel is unbalanced in the sense that the manufacturing and automobile sub-samples differ in size (224 vs. 154 observations). Third, as the data are sourced from listed firms, survivorship bias may understate the true incidence of distress among the broader MSME population. Fourth, the study does not employ firm fixed effects due to the relatively short panel and the focus on between-firm variation.

Chapter 4: Data Analysis

4.1 Descriptive Statistics

4.1.1 Manufacturing Sector

The manufacturing sub-sample comprises 32 firms and 224 firm-year observations over FY2019–FY2025. Table 1 presents descriptive statistics for the key variables.

Variable	Mean	Std. Dev.	Min	Median	Max	Count
Current Ratio	2.33	2.14	0.19	1.78	19.18	224
DIO (Inventory Days)	101.50	87.19	0.00	78.65	527.00	224
DSO (Days Receivable)	114.87	119.29	0.00	81.68	833.00	224
DPO (Days Payable)	174.03	267.79	0.00	93.17	3,772.00	224
Size (ln Assets)	5.08	0.69	3.03	5.21	6.52	224
Leverage	0.17	0.16	0.00	0.15	0.89	224
ROA	0.09	0.07	−0.18	0.09	0.29	224
Distress Dummy (=1)	0.112	0.316	0	0	1	224

Table 1: Descriptive Statistics — Manufacturing Sector

The mean current ratio of 2.33 with a standard deviation of 2.14 indicates considerable heterogeneity in liquidity positions. DSO (mean: 114.87 days) substantially exceeds DIO (mean: 101.50 days), consistent with the hypothesis that receivables are the primary working-capital concern in manufacturing. DPO exhibits the widest dispersion (standard deviation: 267.79 days), reflecting heterogeneous supplier payment practices. Notably, 11.2 per cent of firm-year observations record current ratios below 1, representing 25 distressed firm-years in the manufacturing sample.

4.1.2 Automobile Sector

The automobile sub-sample comprises 22 firms and 154 firm-year observations. Table 2 presents the corresponding descriptive statistics.

Variable	Mean	Std. Dev.	Min	Median	Max	Count
Current Ratio	2.20	1.83	0.28	1.56	13.84	154
DIO (Inventory Days)	71.62	50.56	0.00	63.80	450.00	154
DSO (Days Receivable)	77.48	57.33	0.00	67.35	446.00	154
DPO (Days Payable)	95.04	72.01	0.00	76.23	454.00	154
Size (ln Assets)	4.82	0.75	1.11	5.05	6.03	154
Leverage	0.18	0.15	0.00	0.16	0.51	154
ROA	0.11	0.07	−0.09	0.11	0.30	154
Distress Dummy (=1)	0.078	0.269	0	0	1	154

Table 2: Descriptive Statistics — Automobile Sector

The automobile sector exhibits a lower mean current ratio (2.20 vs. 2.33) and a lower distress incidence (7.8 per cent, or 12 distressed observations). DIO (mean: 71.62 days) and DSO (mean: 77.48 days) are more closely balanced in the automobile sector than in manufacturing. ROA is marginally higher in the automobile sector (0.11 vs. 0.09), consistent with the generally tighter operational disciplines in supply-chain-oriented auto clusters.

4.2 Comparative Sectoral Analysis

Metric	Manufacturing	Automobile	Difference
Current Ratio (mean)	2.33	2.20	+0.13
DIO — Inventory Days	101.50	71.62	+29.88*
DSO — Days Receivable	114.87	77.48	+37.39*
DPO — Days Payable	174.03	95.04	+78.99*
Cash Conversion Cycle (mean)	42.3 days	54.1 days	−11.8
Leverage	0.17	0.18	−0.01
ROA	0.09	0.11	−0.02
Distress Incidence	11.2%	7.8%	+3.4%

Table 3: Comparative Mean Metrics — Manufacturing vs. Automobile

The cross-sector comparison reveals pronounced structural differences. Manufacturing firms maintain significantly higher DSO (114.87 vs. 77.48 days) and DPO (174.03 vs. 95.04 days). The higher DPO in manufacturing suggests that these firms negotiate extended payment terms from suppliers, using trade payables as a principal source of working-capital financing. The higher distress incidence in manufacturing (11.2 vs. 7.8 per cent) despite a comparable current ratio suggests that liquidity risk is more volatile and episodic in manufacturing, concentrated in specific firms and periods.

A noteworthy finding is the difference in the Cash Conversion Cycle (CCC). Manufacturing firms record a mean CCC of 42.3 days, while automobile firms record 54.1 days. The lower CCC in manufacturing is driven principally by the much higher DPO (trade payables extended by suppliers), which offsets the elevated DSO and DIO. This indicates that manufacturing MSMEs, while exposed to elevated receivables risk, are also more aggressive in using supplier credit as a financing buffer.

4.3 T-Test for DSO Differences Between Sectors

Statistic	Manufacturing DSO	Automobile DSO
Mean	114.87	77.48
Variance	14,229.69	3,286.65
Observations	224	154
Hypothesised Mean Difference	0	—
Degrees of Freedom (df)	342	—
t Statistic	3.996	—
P(T≤t) two-tail	0.000080	—
t Critical (two-tail, α=0.05)	1.967	—

Table 4: T-Test for DSO Differences Between Sectors

The t-statistic of 3.996 (p = 0.000080, two-tailed) strongly rejects the null hypothesis of equal mean DSO across sectors. Manufacturing firms collect receivables significantly more slowly than automobile firms, corroborating the structural argument that receivables exposure is a distinctive feature of manufacturing liquidity risk.

4.4 Correlation Analysis

The correlation matrix for the manufacturing sub-sample reveals that leverage shows the strongest negative correlation with Current Ratio (−0.40), confirming that more indebted firms tend to carry lower liquidity buffers. ROA shows negative correlations with DIO (−0.28) and DSO (−0.33), suggesting that firms with longer working-capital cycles are less profitable. DSO and DPO are moderately correlated (0.36), reflecting the common tendency for firms with slower collections to also delay payments to suppliers. The correlations among the three working-capital variables (DIO, DSO, DPO) are generally below 0.40, indicating that multicollinearity is unlikely to severely distort regression coefficients.

4.5 OLS Regression Results: Manufacturing Sector (Binary Distress)

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Sig.
Intercept	0.3993	0.1548	2.578	0.0106	*
Current Ratio	−0.0216	0.0089	−2.428	0.0160	*
DIO (Inventory Days)	−0.000312	0.000206	−1.513	0.1316	n.s.
DSO (Days Receivable)	−0.000332	0.000171	−1.941	0.0536	.
DPO (Days Payable)	+0.000168	0.000066	+2.543	0.0117	*
Size (ln Assets)	−0.009462	0.026596	−0.356	0.7220	n.s.
Leverage	+0.185181	0.134282	+1.379	0.1693	n.s.
ROA	−2.0796	0.2661	−7.821	<0.0001	***
R-squared	0.323	—	—	—	—

Adj. R-squared	0.298	—	—	—	—
F-statistic	14.53	—	—	<0.0001	***
Observations (n)	224	—	—	—	—

Table 5: OLS Regression — Manufacturing Sector (Binary Distress Dummy)

*Note: *p < 0.05, **p < 0.01, ***p < 0.001; . p < 0.10; n.s. = not significant*

The manufacturing regression explains 32.3 per cent of variation in the distress dummy (adjusted $R^2 = 0.298$). Key findings: DSO is a borderline-significant predictor of distress ($p = 0.054$), providing partial support for H_0. DIO is not significant ($p = 0.132$), confirming that inventory management is not the primary trigger of acute distress events in manufacturing. DPO is positively and significantly associated with distress ($p = 0.012$), indicating that firms delaying supplier payments are more likely to be in distress — likely a symptom of cash-flow pressure rather than a cause. ROA is the strongest predictor overall ($p < 0.001$): more profitable firms are substantially less likely to be in distress.

4.6 OLS Regression Results: Automobile Sector (Binary Distress)

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Sig.
Intercept	0.3173	0.1412	2.247	0.0261	*
Current Ratio	−0.0056	0.0154	−0.363	0.7169	n.s.
DIO (Inventory Days)	−0.001197	0.000636	−1.881	0.0620	.
DSO (Days Receivable)	−0.000156	0.000420	−0.371	0.7112	n.s.
DPO (Days Payable)	+0.001262	0.000397	+3.178	0.0018	**
Size (ln Assets)	−0.049986	0.027798	−1.798	0.0742	.
Leverage	+0.423858	0.169220	+2.505	0.0134	*
ROA	−0.764971	0.342439	−2.235	0.0270	*
R-squared	0.233	—	—	—	—
Adj. R-squared	0.191	—	—	—	—
F-statistic	6.50	—	—	<0.0001	***
Observations (n)	154	—	—	—	—

Table 6: OLS Regression — Automobile Sector (Binary Distress Dummy)

*Note: *p < 0.05, **p < 0.01, ***p < 0.001; . p < 0.10; n.s. = not significant*

In the automobile binary specification, DIO carries a negative coefficient with borderline significance ($p = 0.062$), directionally consistent with H_1 but not attaining the 5% threshold. DPO is the only strongly significant working-capital variable ($p = 0.002$). ROA remains a robust negative predictor ($p = 0.027$). The somewhat weaker model fit (Adj. $R^2 = 0.191$) compared to manufacturing reflects greater heterogeneity in the automobile distress sample.

4.7 Robustness: OLS Regressions with Current Ratio (Continuous)

4.7.1 Manufacturing Sector

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Sig.
Intercept	5.8067	1.1148	5.204	<0.0001	***
DIO (Inventory Days)	0.002009	0.001569	1.281	0.2015	n.s.
DSO (Days Receivable)	0.006893	0.001219	5.655	<0.0001	***
DPO (Days Payable)	−0.001231	0.000497	−2.480	0.0139	*
Size (ln Assets)	−0.674653	0.197758	−3.412	0.0008	***
Leverage	−5.115373	0.965167	−5.300	<0.0001	***
R-squared	0.275	—	—	—	—
Adj. R-squared	0.252	—	—	—	—
Observations (n)	224	—	—	—	—

Table 7: OLS Regression — Manufacturing Sector (Current Ratio as Dependent Variable)

An important and striking finding emerges from the continuous specification for manufacturing: DSO becomes highly significant and positive (coefficient: +0.0069; $p < 0.001$). Manufacturing firms with longer debtor days tend to carry higher current ratios. This may appear counterintuitive but reflects the accounting reality that trade receivables — a current asset — mechanically inflate the current ratio numerator. Firms that extend more credit to customers accumulate receivables (current assets), which boosts the current ratio even as cash-flow risk rises. DIO is not significant ($p = 0.201$) in this specification. Leverage is the dominant control variable ($p < 0.001$), and firm size is also significant ($p < 0.001$).

4.7.2 Automobile Sector

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Sig.
Intercept	1.7739	0.7400	2.396	0.0178	*
DIO (Inventory Days)	0.008516	0.003330	2.558	0.0115	*
DSO (Days Receivable)	0.004186	0.002219	1.886	0.0613	.
DPO (Days Payable)	−0.004785	0.002086	−2.295	0.0232	*
Size (ln Assets)	0.226932	0.147349	1.540	0.1257	n.s.
Leverage	−6.509174	0.727823	−8.944	<0.0001	***
R-squared	0.412	—	—	—	—
Adj. R-squared	0.383	—	—	—	—
Observations (n)	154	—	—	—	—

Table 8: OLS Regression — Automobile Sector (Current Ratio as Dependent Variable)

The continuous specification for the automobile sector provides the clearest and most robust finding of the study. DIO is a highly significant positive predictor of the current ratio (coefficient: +0.0085; $p = 0.012$). Automobile firms with longer inventory days carry higher current ratios. While this direction appears counterintuitive at first, it reflects the accounting identity: inventory (a current asset) contributes to the

numerator of the current ratio. Critically, DIO is the dominant working-capital predictor in this specification, confirming H_1 that inventory management is the central liquidity driver in automobile MSMEs. The adjusted R^2 of 0.383 indicates a good model fit, exceeding the binary distress models. Leverage is a particularly strong negative predictor (−6.51; $p < 0.001$).

4.8 Pooled Interaction Regression

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Sig.
Intercept	0.2903	0.1078	2.694	0.0074	**
DIO (Inventory Days)	−0.000289	0.000201	−1.440	0.1507	n.s.
DSO (Days Receivable)	−0.000413	0.000147	−2.819	0.0051	**
DPO (Days Payable)	+0.000207	0.000062	+3.323	0.0010	***
Size (ln Assets)	−0.010843	0.019224	−0.564	0.5728	n.s.
Leverage	+0.387193	0.093952	+4.122	<0.0001	***
ROA	−1.7116	0.21090	−8.111	<0.0001	***
Sector Dummy	+0.031781	0.052625	+0.604	0.5465	n.s.
DIO × Sector Dummy	−0.000566	0.000567	−0.998	0.3188	n.s.
R-squared	0.248	—	—	—	—
Adj. R-squared	0.229	—	—	—	—
Observations (n)	378	—	—	—	—

Table 9: Pooled Interaction Regression — Dependent Variable: Distress Dummy

The pooled interaction term DIO × SectorDummy does not attain statistical significance ($p = 0.319$). The sector dummy itself is also not significant ($p = 0.547$). These results suggest that while directional sector-specific patterns exist in the sub-sample regressions, the cross-sectoral difference in DIO's marginal effect on distress is not statistically distinguishable in a pooled specification. DSO remains a significant predictor in the pooled model ($p = 0.005$), and ROA remains the strongest overall control ($p < 0.001$).

4.9 Summary of Hypothesis Testing

Hypothesis	Expected Finding	Binary Result	Continuous Result	Supported?
H_0: DSO is primary distress driver in Manufacturing	DSO significant in Mfg	DSO borderline sig. (p=0.054)	DSO highly sig. (p<0.001)	Partially
H_1: DIO is primary distress driver in Automobile	DIO significant in Auto	DIO borderline (p=0.062)	DIO highly sig. (p=0.012)	Yes (robustness)
H_2: DIO effect differs across sectors (interaction)	Interaction term significant	Not sig. (p=0.319)	—	Not Supported

Table 10: Summary of Hypothesis Testing Outcomes

Chapter 5: Analysis

5.1 Overview

The empirical findings presented in Chapter 4 offer a nuanced picture of liquidity management in Indian MSMEs. While both manufacturing and automobile sectors face shared liquidity pressures — elevated leverage, lower profitability, and working-capital constraints — the specific working-capital component that most powerfully influences liquidity differs directionally between sectors, as hypothesised. This chapter interprets these findings, relates them to the extant literature, and discusses their managerial and policy implications, along with the limitations of the study.

5.2 Manufacturing Sector: The Centrality of Receivables

The borderline significance of DSO in the binary distress specification ($p = 0.054$), combined with its strong significance in the continuous current-ratio regression ($p < 0.001$), establishes receivables as the most distinctive working-capital dimension of manufacturing liquidity risk. Manufacturing firms — particularly those producing industrial equipment, engineering components, and specialised goods — typically sell on credit to a concentrated base of industrial buyers, resulting in elevated receivable cycles. The mean DSO of 114.87 days in the manufacturing sample — nearly four months of revenue outstanding — reflects this structural reality.

The significant positive coefficient on DSO in the current-ratio regression (coefficient: +0.0069) initially appears paradoxical. However, it is resolved by recognising that trade receivables are a current asset: firms extending more credit accumulate receivables that mechanically inflate the current ratio numerator. This creates a “false liquidity” effect — the current ratio may look healthy even as cash-flow risk rises, because outstanding receivables are booked as liquid assets when they may in practice be slow-moving or at risk of default. This finding aligns with Bardia (2004), who highlighted receivables mismanagement as a key liquidity risk in smaller Indian firms, and with the observations of Chandrabai and Rao (2011) that debtor conversion periods dominate the operating cycle in capital goods manufacturers.

The significant positive coefficient on DPO in both manufacturing specifications also merits interpretation. Higher DPO — longer payment terms to suppliers — is associated with higher distress probability in the binary model. This apparent contradiction with the standard “DPO as financing” narrative is resolved by recognising that in the binary distress model, very high DPO may be a symptom of a firm under payment stress (extending payables involuntarily to preserve cash), while in the current ratio model, higher payables mechanically reduce current liabilities, improving the ratio. This finding underscores the importance of distinguishing between voluntary and involuntary payables extension in MSME analysis.

5.3 Automobile Sector: The Role of Inventory

In the automobile sector, DIO (inventory days) emerges as the dominant working-capital variable in the continuous liquidity specification ($p = 0.012$), confirming H_1. The positive direction of this coefficient reflects the accounting reality that inventory, as a current asset, contributes to the numerator of the current ratio. However, this inventory-driven liquidity buffer is potentially deceptive — slow-moving or excess inventory represents locked capital that may not be readily convertible to cash, particularly in volatile OEM demand environments.

The finding that DIO drives liquidity in automobile MSMEs is consistent with the industry’s operational characteristics. Multi-tier automotive supply chains require component manufacturers to maintain strategic buffer stocks against just-in-time (JIT) demand variability, OEM scheduling changes, and logistics disruptions. The mean DIO of 71.62 days — over two months of production cost tied up in inventory — reflects this structural reality. Firms that allow inventory days to rise beyond operational necessity risk both cash conversion inefficiency and the risk of obsolescence as vehicle models are updated.

The borderline significance of DIO in the binary distress model ($p = 0.062$) is directionally consistent with H_1 but does not attain the conventional 5% threshold. This suggests that high inventory, while a liquidity management challenge, does not necessarily precipitate acute distress events (current ratio dropping below 1) in the automobile sector, at least not at the levels of statistical power available in a 154-observation sub-sample. Instead, inventory management shapes the overall liquidity buffer level, as captured by the continuous current-ratio specification.

5.4 Cross-Cutting Findings

Across both sectors, ROA is the most consistently powerful predictor of distress: profitable firms carry substantially lower distress risk. This finding underscores the well-documented profitability-liquidity nexus and suggests that improving earnings is a more powerful liquidity shield than any individual working-capital improvement. Leverage is also consistently significant, confirming that more indebted firms face higher distress risk regardless of sector.

The finding that the pooled interaction term is not statistically significant does not necessarily invalidate the sectoral narrative. The sub-sample regressions show consistent directional patterns: DSO is borderline in manufacturing binary (and strongly significant in the continuous model), while DIO is borderline in automobile binary (and strongly significant in continuous). The statistical power issue — particularly for the 154-observation automobile sub-sample — and the non-normal distribution of working-capital variables may both attenuate the interaction estimate. The directional evidence from sector-specific regressions provides substantive support for the hypothesis of sectoral heterogeneity even where formal pooled tests fall short.

5.5 Comparison with Prior Literature

The findings are broadly consistent with the broader Indian empirical literature. Patta (2023) found that manufacturing companies showed higher DSO and leverage but also superior ROA and ROE, consistent with the present study's finding that manufacturing MSMEs operate with longer receivable cycles but compensate through higher profitability and aggressive use of supplier credit. The working capital study of ITC Limited by Shroff (2013) found that ROTA was not highly sensitive to working capital leverage, suggesting that sector-specific asset structures mediate the working capital–performance relationship — a finding consistent with the present study's evidence that different sectors have different dominant liquidity levers.

The cement sector study by Alrjoub et al. (2012) found significant inter-firm differences in inventory size and receivables among Rajasthan cement companies, reinforcing the principle that granular, firm-level analysis within sectors reveals variation that aggregate comparisons obscure. The electrical equipment study by Chandrabai and Rao (2011) found that debtor conversion periods dominated the operating cycle in BHEL, while inventory conversion periods were much longer in ABB — a finding that directly parallels the present study's sector-level difference in dominant working-capital drivers.

5.6 Limitations

Several limitations of the study should be acknowledged. First, the sample is restricted to listed MSMEs, which are more transparent and financially stronger than the broader MSME population; findings may not generalise to unlisted or informal-sector firms. Second, the study uses OLS on a binary dependent variable; logit or probit models would handle the binary outcome more rigorously. Third, the absence of firm fixed effects means that time-invariant unobserved heterogeneity is not fully controlled. Fourth, the study covers 54 firms across two sectors; a larger, more diversified sample would provide greater statistical power for interaction tests. Fifth, winsorisation at the 1st/99th percentile, while standard practice, results in some information loss from extreme observations.

Chapter 6: Conclusions and Recommendations

6.1 Summary of Findings

This dissertation set out to examine the determinants of short-term liquidity distress in Indian MSMEs and to assess whether the dominant working-capital driver differs between manufacturing and automobile sectors. Using panel data from 54 listed MSME firms over FY2019–FY2025 (378 observations), sector-wise OLS regressions, robustness checks, and a pooled interaction model, the study arrives at the following principal conclusions:

- DSO (receivables days) is a borderline-significant predictor of binary liquidity distress in manufacturing MSMEs ($p = 0.054$) and a strongly significant predictor of the continuous current-ratio measure ($p < 0.001$), partially confirming H_0. Receivables management is the most distinctive working-capital lever for manufacturing liquidity.
- DIO is a significant predictor of the current ratio (continuous liquidity) in automobile MSMEs ($p = 0.012$), confirming H_1 under the robustness specification. Inventory management is the central working-capital lever for automobile firms' liquidity positions.
- ROA is the single most powerful predictor of distress probability across both sectors, emphasising that profitability provides the primary buffer against short-term liquidity risk.
- The pooled interaction term does not attain significance, suggesting that the sectoral differences in DIO's marginal effect, while directionally consistent, are not statistically distinguishable in the pooled model after controlling for firm characteristics.
- Manufacturing MSMEs exhibit higher distress incidence (11.2 vs. 7.8 per cent), higher DSO (114.87 vs. 77.48 days), and substantially higher DPO (174.03 vs. 95.04 days), while automobile MSMEs show a lower CCC (54.1 vs. 42.3 days) and are more sensitive to inventory fluctuations.

6.2 Managerial Recommendations

6.2.1 Recommendations for Manufacturing MSMEs

Manufacturing MSMEs should treat receivables management as the highest-priority working-capital initiative. The mean DSO of 114.87 days in the sample represents a substantial and manageable source of liquidity risk. The following specific actions are recommended:

Implement a tiered debtor monitoring system. Manufacturing firms should classify all outstanding receivables by age: current (0–30 days), overdue (31–60 days), at-risk (61–90 days), and critical (>90 days). Each tier should trigger a predefined escalation response — automated reminder at 30 days, account manager call at 45 days, formal demand notice at 60 days, and legal/TReDS referral beyond 90 days. This structured approach replaces the informal, relationship-dependent collection practices that tend to prevail in small firms and that contribute to DSO creep. A realistic medium-term target for DSO reduction would be 75–85 days, achievable over a 12–18 month period through consistent application of this framework.

Leverage the TReDS (Trade Receivables Discounting System) platform systematically. TReDS allows an MSME supplier to upload a trade receivable confirmed by a large corporate buyer and receive immediate financing from a registered financier at a competitively tendered rate. The key operating insight is that TReDS converts a 90–120 day receivable into a same-day or next-day cash inflow, reducing effective DSO to near zero for eligible receivables. Many manufacturing MSMEs remain unregistered on TReDS due to unfamiliarity with the platform, concerns about processing fees, or reluctance to involve third-party financiers in buyer relationships. These barriers are surmountable: TReDS fees typically range from 0.50–1.50 per cent per annum (annualised), which is far below the implicit cost of carrying the receivable on the balance sheet (opportunity cost of capital, risk of default, administrative collection burden). MSMEs should target registering their top 5–10 buyers on TReDS as an immediate priority.

Diversify the buyer base to reduce concentration risk. A significant source of DSO vulnerability in manufacturing MSMEs is concentration — situations where two or three large buyers account for 70–80 per cent of revenue. Under these conditions, payment delays by any one buyer create disproportionate liquidity stress. MSMEs should set an explicit revenue concentration target: no single buyer should account for more than 30–35 per cent of total revenue. While diversification takes time, even partial progress — adding two to three new industrial customers per year — meaningfully reduces the receivables-related tail risk.

Restructure payment terms in new contracts to incorporate milestone-linked settlements. Rather than accepting single end-of-contract payment terms, manufacturing MSMEs should negotiate staggered payment schedules: for example, 30 per cent on order placement, 40 per cent on delivery, and 30 per cent within 30 days of acceptance. This structure shortens the effective collection cycle without requiring buyers to change their banking timelines. It is particularly feasible for project-based or custom manufacturing engagements where deliverables are clearly milestoned.

Establish a minimum cash buffer equal to 45–60 days of fixed operating expenses. The distress analysis shows that 11.2 per cent of manufacturing firm-year observations fall below a current ratio of 1.0 — an acute vulnerability that typically develops over quarters rather than overnight. Maintaining a liquidity reserve equivalent to one to two months of rent, payroll, and debt service provides a buffer that allows the business to continue operating while receivables are collected or alternative financing is arranged. This buffer should be held in a liquid instrument (current account or short-term FD) and should not be confused with working capital deployed in operations.

Audit DPO extension practices annually to distinguish strategic from distressed payables stretching. The positive association between DPO and distress in the manufacturing regression ($p = 0.012$) reflects that some firms in the sample are involuntarily extending payables because they cannot pay on time, rather than as a deliberate financing strategy. Finance teams should distinguish between these two situations: planned payable extensions negotiated upfront with suppliers are healthy; reactive delays driven by cash-flow shortfalls are a warning signal. An annual DPO audit — comparing actual payment timelines against contracted payment terms — provides an early indicator of developing liquidity stress.

6.2.2 Recommendations for Automobile MSMEs

Automobile and auto-component MSMEs should treat inventory management as their primary working-capital priority. With a mean DIO of 71.62 days and strong empirical evidence that DIO shapes the current-ratio level, inventory decisions are the most controllable and impactful lever available to auto-component CFOs.

Conduct an SKU-level inventory audit and implement ABC classification. Not all inventory is created equal: a small number of high-value, high-velocity SKUs (Class A) account for the majority of inventory value and should be managed with tight reorder discipline, while a large number of low-value, slow-moving items (Class C) represent the greatest risk of capital lock-up and obsolescence. Auto-component firms that implement ABC classification and apply different holding norms to each class typically achieve DIO reductions of 10–15 days within one to two production cycles. A realistic medium-term DIO target for automobile MSMEs would be 55–60 days, achievable through structured inventory reduction across Class B and C items.

Engage OEM customers in demand-visibility sharing arrangements. A significant driver of excess inventory in auto-component firms is demand uncertainty: firms stock buffer inventory to guard against OEM schedule changes or rush orders. However, many Tier-1 OEMs now share 8–12 week rolling production forecasts through supplier portals, which can substantially reduce the safety-stock requirement. Auto-component MSMEs that are not actively accessing and operationalising these forecasts are leaving a major inventory-reduction opportunity unexploited. Finance managers should work with production and supply-chain teams to establish a formal process for translating OEM forecast data into reorder point adjustments on a monthly basis.

Explore Vendor-Managed Inventory (VMI) arrangements for raw materials procurement. Under a VMI arrangement, the supplier of raw materials (for example, steel, aluminium, or rubber) takes responsibility for

replenishing inventory at the auto-component firm's facility, based on agreed minimum and maximum stock levels. This transfers the working-capital burden of raw material stockholding to the supplier, reducing the auto-component firm's DIO for raw materials to near zero. VMI is increasingly common in the automotive supply chain and is worth negotiating with key raw material suppliers, particularly those who supply multiple auto-component firms in the same geographic cluster.

Set and monitor monthly DIO targets by product category. A common weakness in MSME financial management is treating DIO as a year-end metric rather than a live operational variable. Finance managers in automobile firms should calculate DIO by product category on a monthly basis and report it alongside revenue and EBIT in the management accounts. Categories where DIO exceeds 90 days should trigger an immediate stock review: is the inventory slow-moving due to demand weakness? Is there a quality issue preventing dispatch? Is there a discrepancy between production planning and actual orders? Monthly DIO monitoring converts a lagging indicator into an early warning system.

Negotiate extended DPO with key raw material suppliers. The automobile regression shows that higher DPO is significantly associated with higher current ratios (coefficient: −0.0048, $p = 0.023$ in the continuous specification, where a higher DPO reduces current liabilities and improves the ratio mechanically). For automobile MSMEs that are financially sound, negotiating extended payment terms of 60–90 days from key suppliers — up from the current mean of 95 days, with some firms on much shorter terms — provides a meaningful implicit financing benefit. This negotiation is most feasible with suppliers who value the long-term relationship and have sufficient financial flexibility to extend terms.

Monitor leverage closely, given its dominant negative effect on current ratios. Leverage is the strongest predictor of current ratio levels in the automobile sector (coefficient: −6.51, $p < 0.001$). Auto-component MSMEs that have taken on substantial term debt to finance equipment, plant expansion, or working capital should model the impact of debt amortisation schedules on their projected current ratios over the next 24–36 months. If the trajectory shows current ratios approaching 1.2–1.3, pre-emptive refinancing or partial prepayment should be considered before the ratio drops into the distress zone. A current-ratio stress test — modelling what happens to liquidity if revenue falls by 15–20 per cent for two consecutive quarters — is a straightforward and valuable exercise that few MSMEs conduct proactively.

6.3 Policy Recommendations

Policymakers and regulatory authorities should consider the following targeted interventions:

- Expansion and deepening of TReDS participation: Making TReDS onboarding mandatory for large corporate buyers in manufacturing-intensive sectors would improve the liquidity of their MSME suppliers.
- Inventory financing schemes for automobile clusters: RBI and SIDBI could introduce specialised inventory funding lines for auto-component MSMEs, providing short-term credit secured against finished goods inventory.
- Payment enforcement mechanisms: Strengthening the enforcement of MSME Act provisions mandating payment within 45 days would reduce structural DSO in the manufacturing sector.
- Financial literacy and working-capital training: MSME development agencies should incorporate working-capital management modules into their enterprise development programmes.
- Targeted credit guarantees: CGTMSE schemes could be redesigned to specifically collateralise receivables and inventory assets, reducing the cost of working-capital financing.

6.4 Scope for Future Research

This study opens several important avenues for future empirical investigation. The following directions are identified as priorities:

1. Panel data with firm fixed effects. The present study uses pooled OLS without firm fixed effects, due to the relatively short panel (seven years) and the focus on cross-sectional variation. A longer panel — ten or more years — would support the inclusion of firm fixed effects, controlling for time-invariant unobserved

heterogeneity (such as management quality, ownership structure, and market position). Fixed-effects estimation would sharpen the identification of the causal effect of working-capital changes on liquidity outcomes.

2. Unlisted MSME data. The present study is restricted to listed MSMEs, which are more transparent and financially stronger than the broader MSME universe. The most financially fragile MSMEs — those most at risk of liquidity distress — are precisely the ones that are not listed. Future research using data from CIBIL MSME Pulse, TReDS transaction records, or GST return data would provide a more representative and policyrelevant picture of MSME liquidity dynamics.

3. Causal identification through policy quasi-experiments. The introduction of TReDS (operationalised from 2017 onwards), the GST e-invoicing mandate (extended to MSMEs in 2021–22), and the Emergency Credit Line Guarantee Scheme (ECLGS) during COVID-19 all represent natural experiments that can be exploited to causally identify the impact of policy interventions on MSME DSO, DIO, and liquidity distress rates. A difference-in-differences design comparing treated firms (those that adopted TReDS or received ECLGS) with matched control firms would produce high-quality causal evidence on policy effectiveness.

4. Machine learning for distress prediction. The three-variable working-capital model (DSO, DIO, DPO) plus controls produces adjusted R^2 values of approximately 0.25–0.40 in the present study. Ensemble machine learning methods (gradient boosting, random forests) trained on a richer feature set — including sector, geography, age of firm, GST filing regularity, and credit bureau data — may substantially improve out-of-sample distress prediction accuracy. Such a model would have direct commercial applications in MSME credit scoring for banks and NBFCs.

5. Qualitative research on working-capital decision-making. The quantitative findings of this study raise questions that quantitative data cannot answer: Why do manufacturing MSMEs accept DSO of 115 days without enforcing their contractual rights? What prevents auto-component firms from implementing ABC inventory classification despite its well-documented benefits? Structured interviews with CFOs and finance managers of MSME firms would provide the “why” behind the patterns identified in the regressions, and would generate practical insights for capability-building interventions.

6. Comparative international study. India’s MSME working-capital context is shaped by specific institutional factors: the MSMED Act, TReDS, GST, and the structure of domestic supply chains. A comparative study of MSME liquidity drivers across India, Vietnam, Bangladesh, and Indonesia — all economies with large, export-oriented MSME sectors — would identify which patterns are universal and which are India-specific, informing the design of globally relevant policy interventions.

6.5 Concluding Remarks

This dissertation provides empirical evidence that the primary working-capital driver of liquidity distress differs between manufacturing and automobile MSMEs in India. In manufacturing, receivables management (DSO) is the critical lever; in automobile firms, inventory management (DIO) is dominant. These findings carry direct implications for MSME financial management and for policymakers designing liquidity-support infrastructure. At a deeper level, the study underscores the importance of sector-context-sensitivity in financial risk analysis: a uniform, sector-agnostic approach to working-capital management will systematically mistarget resources and leave the most acute vulnerabilities unaddressed. As India’s MSME sector continues to evolve — expanding into global supply chains, adopting digital technologies, and confronting new demand volatility — the ability to manage sector-specific liquidity risks with precision will become increasingly central to the sector’s long-term resilience and competitiveness.

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Appendices

Appendix A: List of Sample Firms

A.1 Manufacturing Sector Firms (32 firms)

Eimco Elecon (India) Ltd., Thejo Engineering Ltd., Lokesh Machines Ltd., Spectrum Electrical Industries Ltd., Batliboi Ltd., Disa India Ltd., ICE Make Refrigeration Ltd., United Drilling Tools Ltd., TRF Ltd., Ducon Technologies Ltd., Mahindra EPC Irrigation Ltd., Emkay Taps & Cutting Tools Ltd., Birla Precision Technologies Ltd., Mazda Ltd., ATV Projects Ltd., Auri Grow India Ltd., Innovators Facade Systems Ltd., Macpower CNC Machines Ltd., BEW Engineering Ltd., Affordable Robotic & Automation Ltd., Josts Engineering Company Ltd., Axtel Industries Ltd., Crown Lifters India Ltd., Stovec Industries Ltd., Hariyana Ship Breakers Ltd., Integra Engineering India Ltd., Taylormade Renewables Ltd., Danlaw Technologies India Ltd., ITL Industries Ltd., D & H India Ltd., Artson Engineering Ltd., Bemco Hydraulics Ltd.

A.2 Automobile/Auto-Component Sector Firms (22 firms)

Wardwizard Innovations & Mobility Ltd., Remsons Industries Ltd., Samkrg Pistons & Rings Ltd., Bimetal Bearings Ltd., Pradeep Metals Ltd., Indag Rubber Ltd., Saint-Gobain Sekurit India Ltd., Veljan Denison Ltd., Akar Auto Industries Ltd., Kinetic Engineering Ltd., Menon Bearings Ltd., Menon Pistons Ltd., Precision Metaliks Ltd., Sintercom India Ltd., Frontier Springs Ltd., Pritika Engineering Components Ltd., Shigan Quantum Technologies Ltd., Simmonds Marshall Ltd., Porwal Auto Components Ltd., G S Auto International Ltd., Bhagwati Autocast Ltd., Duncan Engineering Ltd.

Appendix B: Variable Definitions Summary

Variable	Definition	Source
Current Ratio	Current Assets / Current Liabilities	Screener / Annual Reports
Distress Dummy	1 if Current Ratio < 1; 0 otherwise	Derived
DIO	Inventory / (COGS/365) — Inventory Days	Screener / Annual Reports
DSO	Trade Receivables / (Revenue/365) — Debtor Days	Screener / Annual Reports
DPO	Trade Payables / (COGS/365) — Creditor Days	Screener / Annual Reports
Size	Natural log of Total Assets (INR Cr.)	Derived
Leverage	Total Debt / Total Assets	Derived
ROA	EBIT / Total Assets	Derived
Sector Dummy	1 = Automobile, 0 = Manufacturing	Derived
DIO × Sector	Interaction: DIO multiplied by Sector Dummy	Derived

Note: All financial variables are winsorised at the 1st and 99th percentile thresholds to reduce outlier influence on regression estimates.